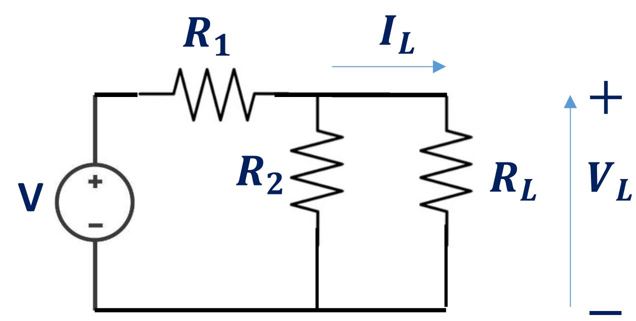
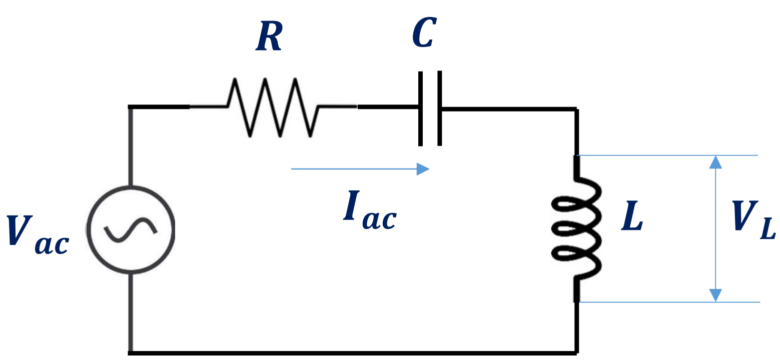
Electric Circuits

Basic Definitions, Voltage & Current Sources, KCL & KVL

Syllabus

DC Analysis: Dependent and independent Voltage and current sources, Nodes, Paths, Loops and Branches, Nodal and Mesh Analysis, Linearity & Superposition Theorem, Source Transformations, Thevenin's and Norton's Theorems, Maximum Power Transfer. RL, RC and RLC Circuit.





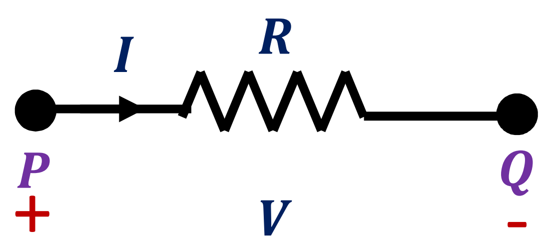
AC Circuit Analysis: Sinusoidal Forcing Function, Phasor Relationship for R, L and C, Impedance and Admittance, Phasor Diagrams. Instantaneous Power, Average Power, Complex Power, Apparent Power and Power Factor.

Tentative Lecture Plan

L	LDay-1	LDay-2	Topics	T	TDay
1	**FRI, 04 AUG**	**MON, 07 AUG**	**Basic Definitions, Voltage & Current Sources, KCL & KVL**	**1**	**THU, 10 AUG**
2	TUE, 08 AUG	FRI, 11 AUG	Nodal Analysis: Independent and Dependent Current Sources	2	SAT, 19 AUG
3	MON, 14 AUG	THU, 17 AUG	Nodal Analysis: Current and Voltage Sources		
4	FRI, 18 AUG	MON, 21 AUG	Mesh Analysis: Voltage and Current Sources	3	THU, 24 AUG
5	TUE, 22 AUG	FRI, 25 AUG	Network Theorems: Superposition Theorem		
			SUN, 27 AUG : Quiz-1 : Lectures 1-5		
6	FRI, 26 AUG	MON, 28 AUG	Network Theorems: Thevenin's and Norton's Theorem	4	THU, 31 AUG
7	TUE, 29 AUG	FRI, 04 SEP	Network Theorems: Maximum Power Transfer Theorem	5	WED, 06 SEP
8	TUE, 05 SEP	FRI, 08 SEP	Series RL and RC Circuits		
9	SAT, 09 SEP	MON, 11 SEP	Series RL, RC, RLC Circuits Response to Constant, Sinusoid and Exponential Input	6	THU, 14 SEP
10	TUE, 12 SEP	FRI, 15 SEP	Concept of Phasor, AC Circuits, AC Power	R	SAT, 16 SEP
			WED, 20 SEP: MidSem : Lectures 1 – 10		

Basic Definitions

- The voltage V between two ends of a path $P \rightarrow Q$ is the total energy required to move a unit electric charge along that path. It is measured in volt and denoted as **V**
- The current I is charge flow through a material in a unit time. It is measured in *ampere* and denoted as **A**
- *Ohm's Law* (An experimental law) - The current I flowing in a circuit is directly proportional to the applied potential difference V and inversely proportional to the resistance R in the circuit. The unit of Resistance is *ohm* and is denoted as Ω. Ohm's law provides a linear relation between V and I : $V = IR$



Superposition Property

$$y = f(x)$$

Additive

$$f(x_1 + x_2) = f(x_1) + f(x_2)$$

Homogeneous

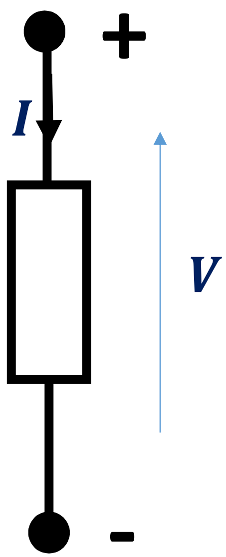
$$f(\alpha x) = \alpha f(x)$$

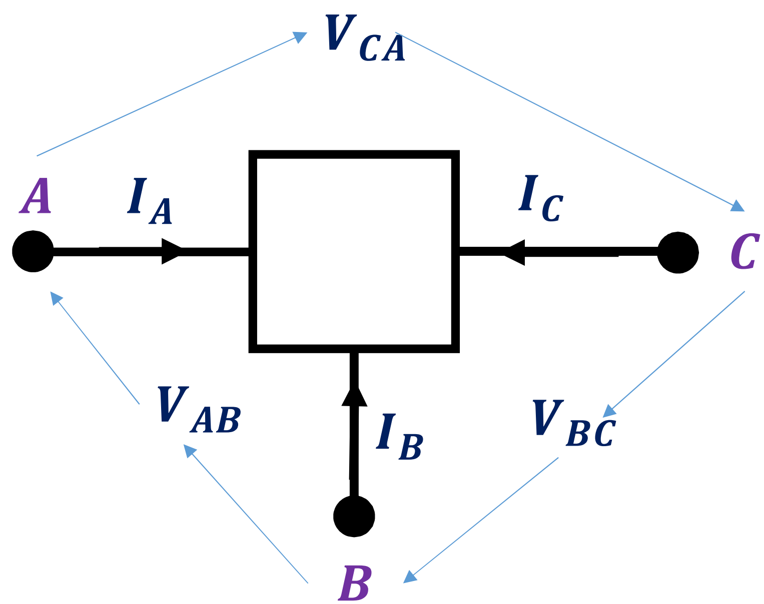
$$f(\alpha x_1 + \beta x_2) = \alpha f(x_1) + \beta f(x_2)$$

Ohm’s Law follows the Superposition Property

Electrical Circuit Element

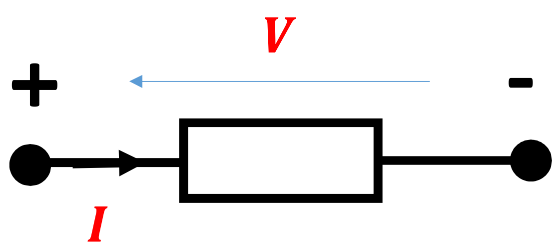
- Electrical Device with at least Two Terminals
- Electric Current can flow In/Out of this device through these terminals
- Electric Potential Difference can be measured across these terminals





Passive Sign Convention

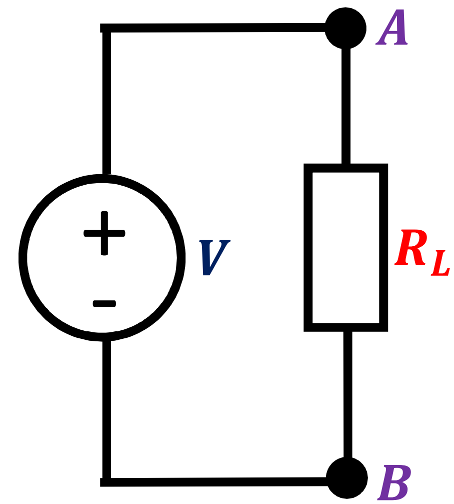
Voltage polarity indicates the relative potential between two points. Here, **+** is assigned to a point with higher potential and **–** is assigned to a lower potential point.

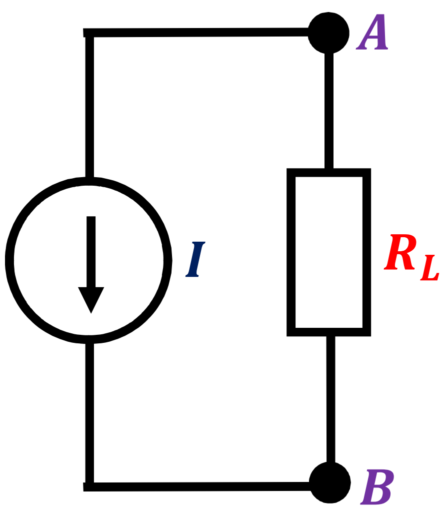


The Current Direction from **+** to **-** is considered as flow of positive charge (physically the opposite direction of the flow of negative charge).

Independent Voltage/Current Sources

Independent Voltage Source: Maintains a Constant Voltage across R_L and is not affected by any other variables.

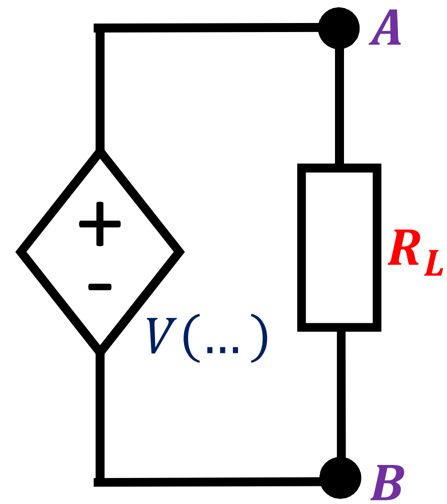


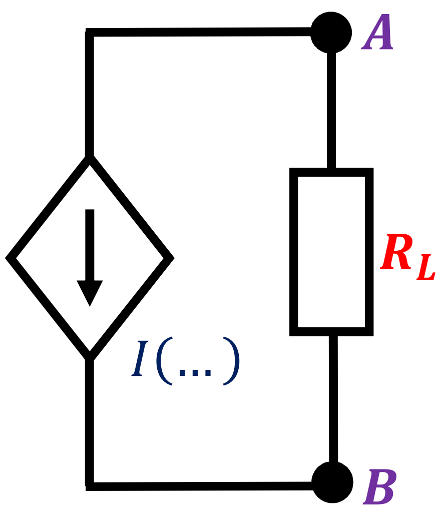


Independent Current Source: Maintains a Constant Current through R_L and is not affected by any other variables.

Dependent Voltage/Current Sources

Dependent Voltage Source: The Voltage across the Source depends on other Variables and is Represented as $V(v_1, v_2, \ldots, i_1, i_2, \ldots)$

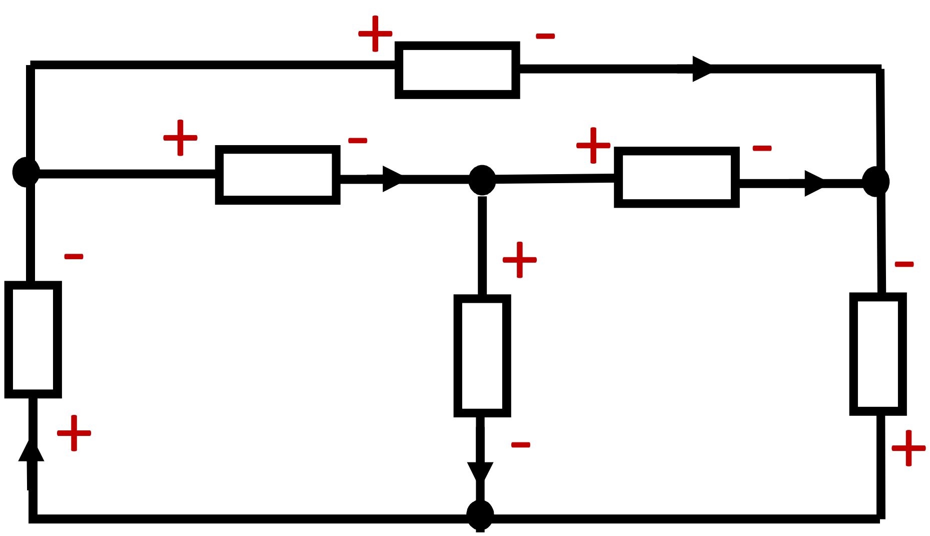




Dependent Current Source: The Current through the Source depends on other Variables and is Represented as $I(v_1, v_2, \ldots, i_1, i_2, \ldots)$

Electrical Circuit

- Interconnected Collection of Electrical Circuit Elements
- Connected such that there is at least one closed loop
- The circuit elements are connected through conducting wires
- The connecting wires converge on *Nodes*
- A *Branch* connects any two *Nodes*
- Each *Branch* hosts a circuit element



Number of Nodes

$n = 4$

Number of Branches

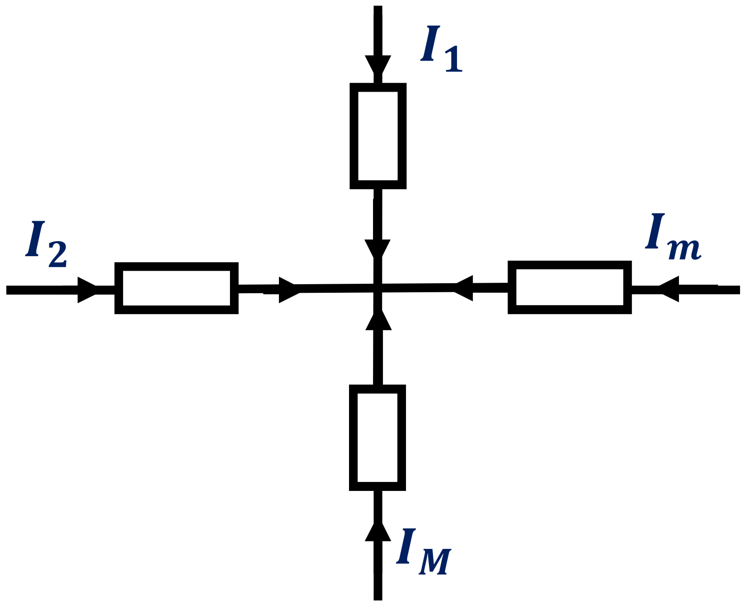
$b = 6$

Electrical Circuit Analysis

- Evaluate all Branch Voltages and Branch Currents of the Electrical Circuit
- A circuit with b branches has $2b$ variables
  - There are b *Branch Voltages*
  - There are b *Branch Currents*
- This requires $2b$ linearly independent equations
- The branch component values contribute up to b equations through Ohm’s law
- The remaining equations are derived from the *interconnection details* of the *circuit topological constraints*

Kirchoff’s Current Law (KCL)

***KCL*:** The algebraic sum of the currents in all branches which converge to a common node is equal to zero

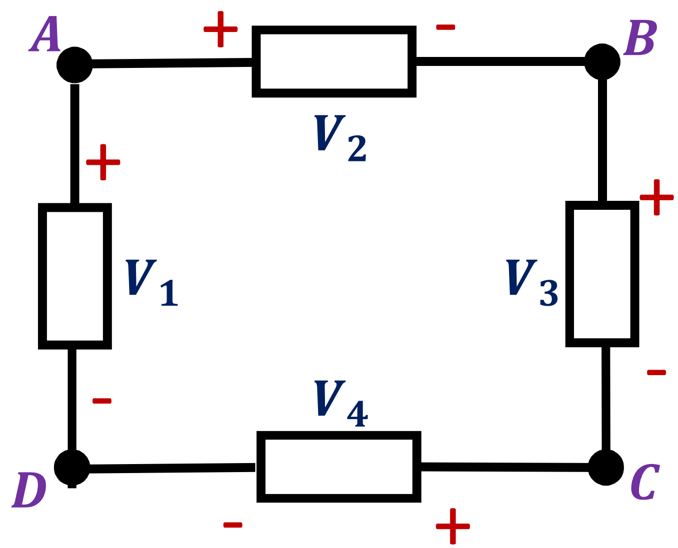


$$\sum_{m=1}^{M} I_m = 0$$

- This can be explained by the principle of conservation of charges
- ***KCL*** provides us with $(n-1)$ linearly independent equations for a circuit with n nodes

Kirchoff’s Voltage Law (KVL)

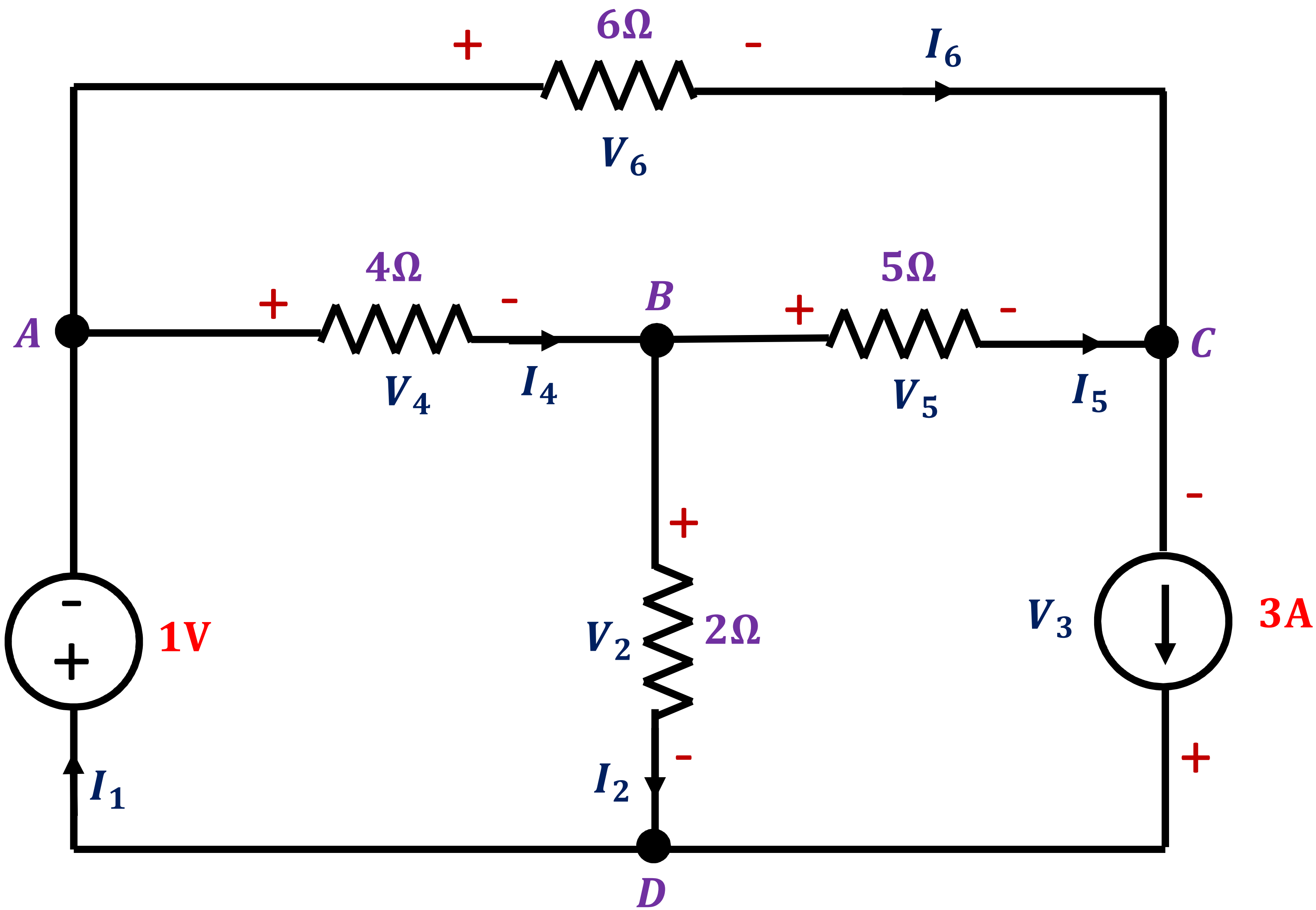
KVL: The algebraic sum of all voltages between successive nodes in a closed path in the circuit is equal to zero

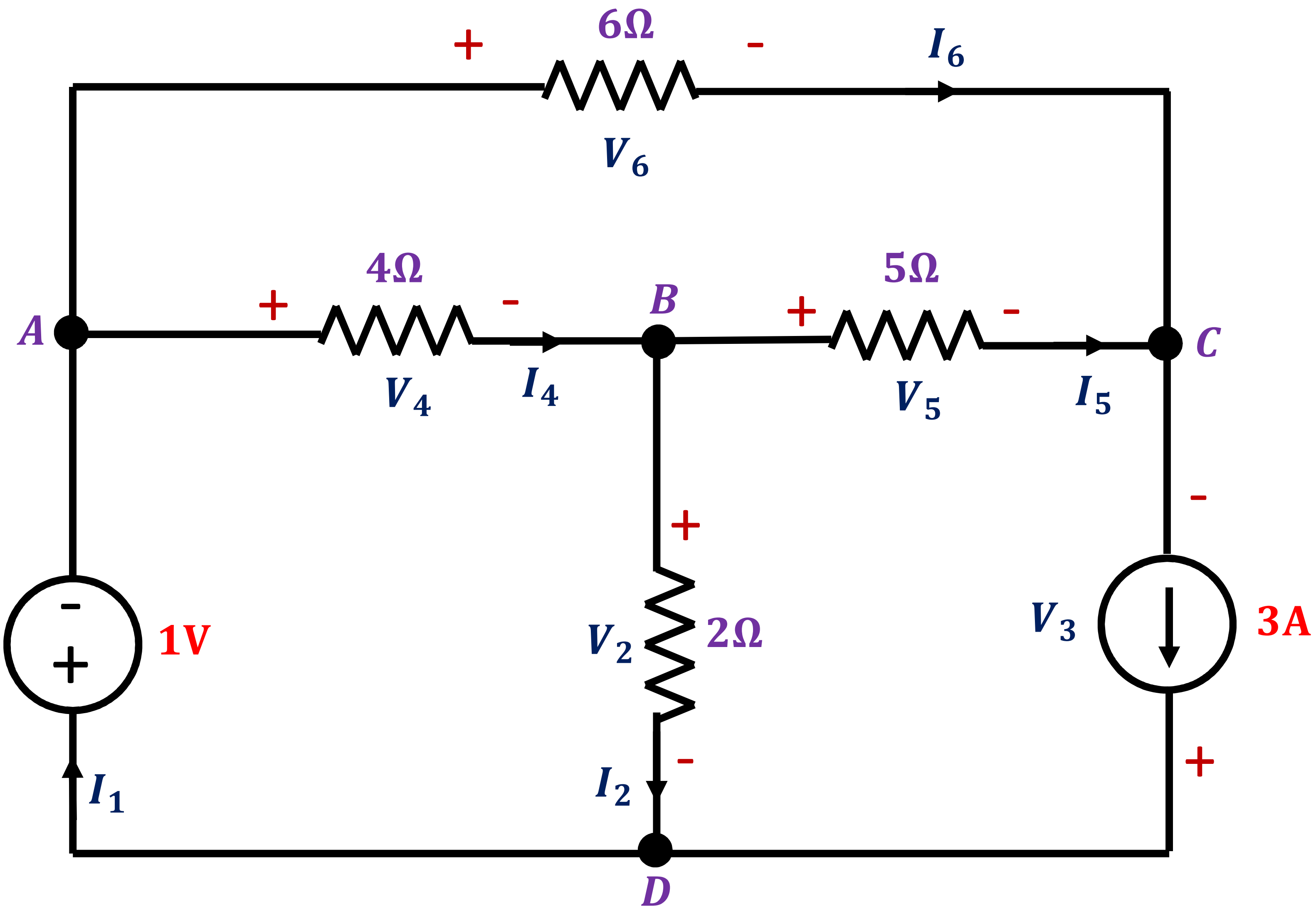


$$A - B - C - D - A$$

$$V_2 + V_3 + V_4 - V_1 = 0$$

- KVL Assumes that the Loops in the Circuit Do Not Enclose any Significant Time Varying Magnetic Field
- ***KVL*** provides us with $\{b - (n-1)\}$ linearly independent equations for a circuit with n nodes and b branches


6Ω
I_6
V_6
4Ω
5Ω
A
B
C
V_4
I_4
V_5
I_5
1V
V_2
2Ω
V_3
3A
I_1
I_2
D



Number of Nodes

$n = 4$

Equations from KCL

$EQ(KCL) = 3$

Number of Branches

$b = 6$

Equations from KVL

$EQ(KVL) = 3$

Solution

- Out of 6 Branch Voltages and 6 Branch Currents, $V_1 = 1V$ and $I_3 = 3A$ are known. This leaves us with 10 variables.
- From Ohm's Law
  - $2I_2 = V_2$; $4I_4 = V_4$; $5I_5 = V_5$; $6I_6 = V_6$
- Using Kirchoff's Current Law
  - Node A: $I_1 = I_4 + I_6$
  - Node B: $I_4 = I_2 + I_5$
  - Node C: $I_5 + I_6 = 3$
- Using Kirchoff's Voltage Law
  - A-B-D-A: $V_4 + V_2 + 1 = 0$
  - B-C-D-B: $V_5 - V_3 - V_2 = 0$
  - A-C-B-A: $V_6 - V_5 - V_4 = 0$

Summary

- Basic Definitions
- Goal of Electrical Circuit Analysis
- Voltage and Current Sources
- Kirchoff’s Current Law (KCL)
- Kirchoff’s Voltage Law (KVL)

Thank You